Project Title: **Development of a Web and Mobile-Based Road Accident Reporting System**
Section: BSIT 3I G1 / BSIT 3I G2
Grouping: Minimum 4 members, Maximum 5 members
Period Timeline: October 12 – 30, 2026
Submission Date: November 02, 2026

1. Project Overview

The Web and Mobile-Based Road Accident Reporting System is designed to provide a convenient and efficient way for the public to report road accidents using a website or mobile application. The system allows users to submit accident details, capture the accident location, upload photos, and monitor the status of their reports.

Authorized administrators and traffic authorities can review reports, verify accident information, manage accident records, and generate reports for monitoring and decision-making.

The primary goal is to improve the speed, accessibility, accuracy, and organization of road accident reporting.

2. Project Objectives

The system aims to:

1. Provide a web-based and mobile-based platform for reporting road accidents.
2. Allow users to report accidents quickly using a digital form.
3. Capture the accident's location through GPS or manual map selection.
4. Allow users to upload accident photos and supporting evidence.
5. Enable authorities to verify, update, and manage accident reports.
6. Allow users to track the status of their submitted reports.
7. Maintain a centralized database of road accident records.
8. Generate statistical reports to help authorities identify accident-prone areas.
9. Improve communication between the public and authorized traffic personnel.
10. Protect accident information through secure authentication and access controls.

3. System Users and Roles

1. Public User / Reporter

- Register and log in to the system, if required.
- Submit road accident reports.
- Provide accident details and location.
- Upload photos or other evidence.
- View personal report history.
- Track report status and receive notifications.

2. Administrator

- Manage user accounts and permissions.
- Review and validate submitted reports.
- Assign reports to authorized personnel.
- Manage accident categories and system settings.
- Monitor system activity.
- Generate and export reports.

3. Traffic Officer / Authorized Personnel

- View assigned accident reports.
- Verify accident information.
- Update investigation and response status.
- Add official remarks and findings.
- Record relevant actions taken.

4. Functional Requirements

Functional requirements describe what the system must do.

ID	Requirement	Description
FR-01	User Registration	Users can create an account using required information.

FR-02	User Login	Users can securely log in and log out.
FR-03	Accident Reporting	Users can submit a report containing accident details.
FR-04	Location Capture	The system can obtain GPS coordinates or accept a manually selected location.
FR-05	Evidence Upload	Users can attach accident photos and permitted supporting files.
FR-06	Report Tracking	Users can view the status of their submitted reports.
FR-07	Report Management	Authorized personnel can view, review, edit, and manage reports.
FR-08	Report Verification	Authorized personnel can verify reports and flag incomplete or duplicate submissions.
FR-09	Notifications	The system can notify users about submission confirmation and status changes.
FR-10	Dashboard	Administrators can view accident statistics and report summaries.
FR-11	Search and Filtering	Personnel can search reports by date, location, status, and accident type.
FR-12	Report Generation	The system can generate daily, weekly, monthly, and annual summaries.
FR-13	User Management	Administrators can manage user accounts and assigned roles.
FR-14	Audit Trail	The system records important administrative actions and report status changes.
FR-15	Data Export	Authorized users can export permitted reports to PDF or CSV.

5. Accident Report Information Requirements

Each accident report should contain the following information.

Field	Description	Required?
Report ID	Unique system-generated identifier	Yes
Reporter Name	Name of the person reporting	Depends on reporting policy
Contact Information	Phone number or email	Depends on reporting policy
Accident Date	Date of the incident	Yes
Accident Time	Approximate or exact time	Yes
Accident Location	Road, street, barangay, municipality, or landmark	Yes
GPS Coordinates	Latitude and longitude, if available	Optional
Accident Type	Collision, hit-and-run, vehicle rollover, pedestrian accident, etc.	Yes
Vehicles Involved	Type and number of vehicles	Yes, where applicable
Injury Information	Whether injuries or fatalities were reported	Yes, including an unknown option
Accident Description	Narrative of what happened	Yes
Photo Evidence	Images of the accident or scene	Optional
Report Status	Current processing status	Yes
Date Submitted	Automatically recorded submission date	Yes
Assigned Officer	Officer responsible for handling the report	If assigned
Official Remarks	Verification notes or investigation findings	Optional

Important: The system should clearly distinguish between a public-submitted report and an officially verified accident record.

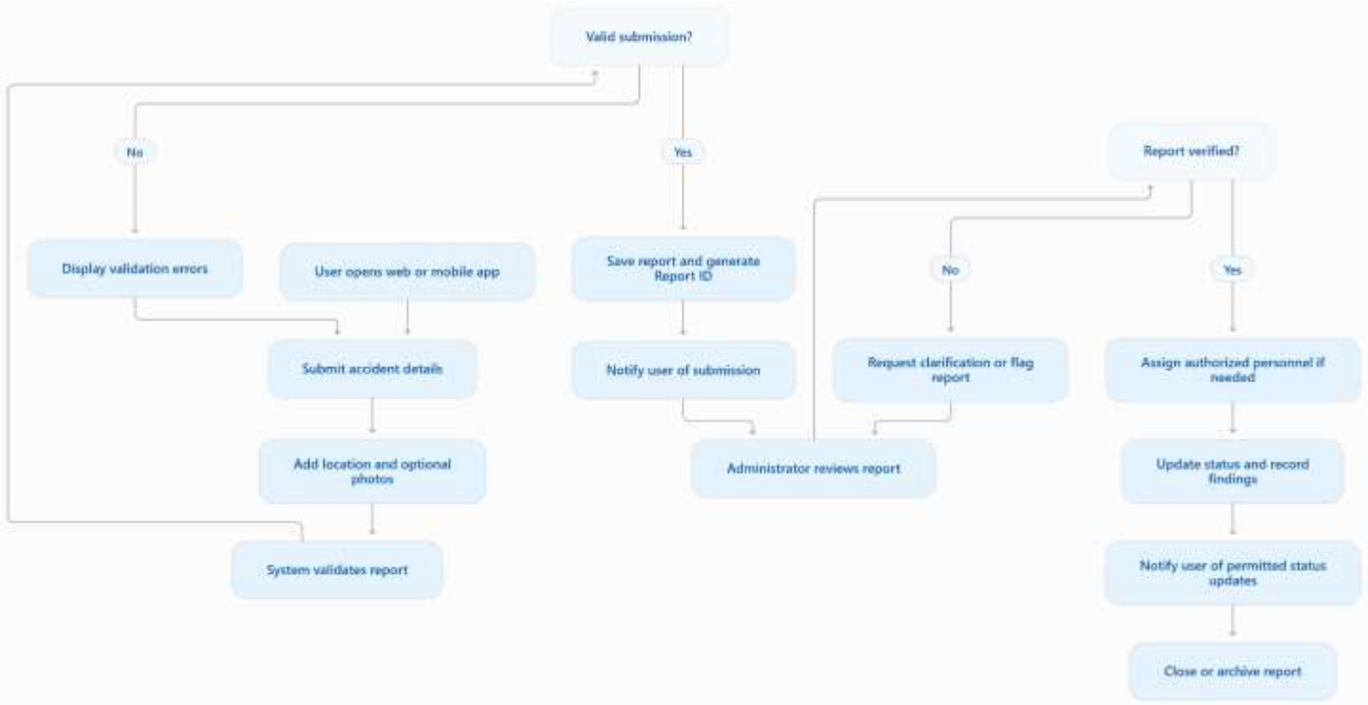
5. Proposed Technology Stack

The following technologies are examples; the final stack will depend on the development team's skills and project budget**.**

Component	Suggested Technology
Web Frontend	HTML, CSS, JavaScript, React, or Bootstrap
Mobile Application	Flutter or React Native
Backend	Node.js with Express, Laravel, or Django
Database	MySQL or PostgreSQL
Maps and GPS	Google Maps Platform, OpenStreetMap, or another mapping provider
Image Storage	Secure server storage or cloud object storage
Notifications	Email, SMS gateway, or mobile push notifications
Authentication	Secure session-based authentication or token-based authentication
Hosting	Cloud hosting or a properly secured institutional server
Version Control	Git and GitHub

A practical student-project option is React or Bootstrap for the web, Flutter for mobile, Laravel for the backend, and MySQL for the database.

6. Accident Report Workflow

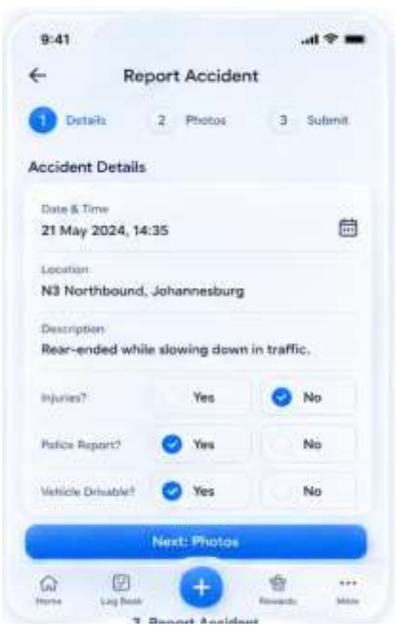


Suggested report statuses:

- Submitted
- Under Review
- Additional Information Required
- Verified
- Assigned
- In Progress
- Resolved / Closed
- Rejected, with a recorded reason

The actual statuses should match the workflow of the responsible traffic authority.

7. Interface Requirements



A. Mobile Application

- Home screen with a prominent “Report an Accident” option.
- Accident details form.
- GPS location and map selection.
- Camera or photo-upload feature.
- Submission confirmation and report tracking.
- Notifications and report history.

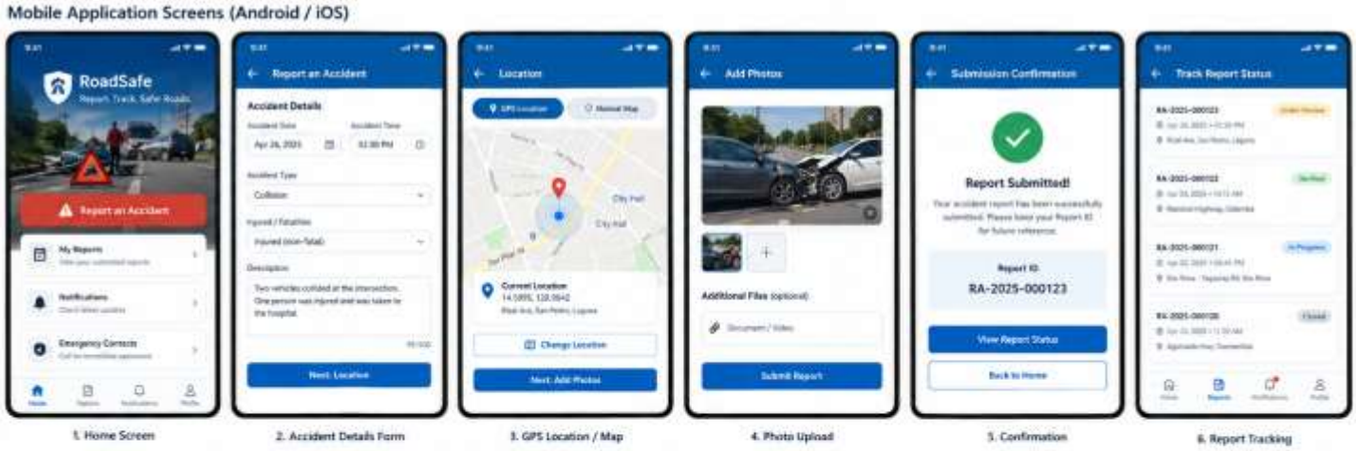


B. Administrator Dashboard

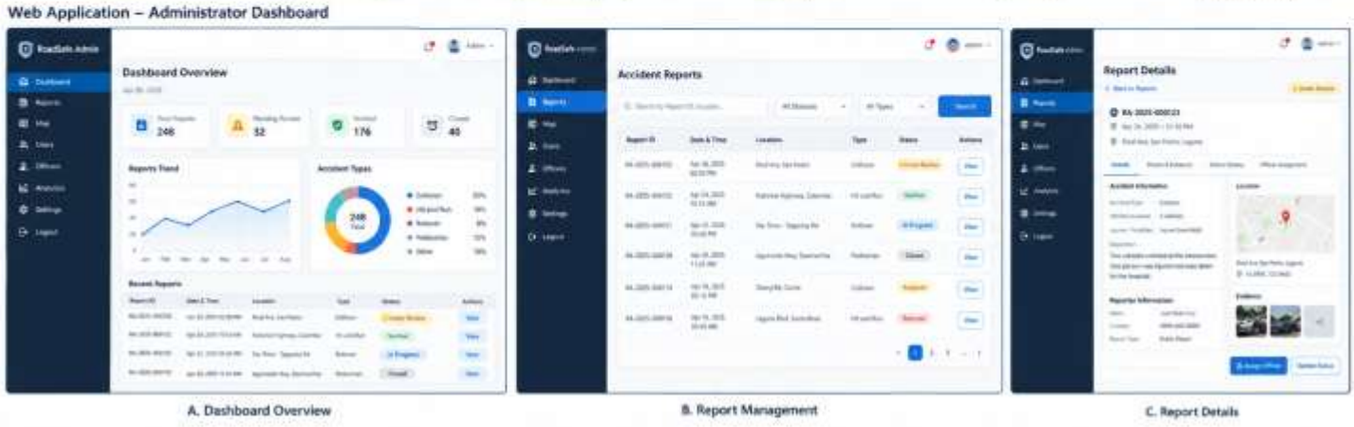
- Summary cards for total, pending, verified, and closed reports.
- Accident report list with filters.
- Map showing reported accident locations.
- Report details and evidence viewer.
- Officer assignment and status management.
- Charts and downloadable reports.

8. Sample Interface

Mobile Application Screens (Android / iOS)



Web Application – Administrator Dashboard



Here is a sample interface design for the Web and Mobile-Based Road Accident Reporting System,

1. Mobile Application Interface



Screen 1: Home Screen

- Report an Accident button
- My Reports
- Notifications
- Emergency Contacts
- Bottom navigation menu



Screen 4: Photo Upload

- Take or upload accident photos
- Preview attached images
- Remove unwanted attachments
- Submit report



Screen 2: Accident Details Form

- Accident date and time
- Accident type
- Injury information
- Description of the incident
- Next button to continue



Screen 5: Submission Confirmation

- Successful submission message
- Unique report ID
- View Report Status button
- Return to home screen



Screen 3: GPS Location

- GPS location detection
- Interactive map
- Manual location selection
- Road and landmark information



Screen 6: Report Tracking

- List of submitted reports
- Report ID and location
- Status indicators
- Date and time of submission

2. Web Application Interface



A. Administrator Dashboard

- Total accident reports
- Pending and verified reports
- Accident statistics and charts
- Recent accident reports
- Accident trends



B. Accident Report Management

- View all submitted reports
- Search by report ID or location
- Filter by date, type, and status
- Review and verify reports
- Assign reports to traffic officers



C. Accident Report Details

- Full accident information
- Map and GPS coordinates
- Uploaded photo evidence
- Reporter information with access restrictions
- Officer assignment and status updates